

Chapter 91 — Renal Transplantation

CHRONIC KIDNEY DISEASE EPIDEMIOLOGY

- **Chronic kidney disease (CKD)**, among the commonest chronic illnesses, affects about **14% of United States (US) adults**, defined by reduced estimated glomerular filtration rate (eGFR), albuminuria, or both (Table 91.1).
- **Kidney Disease: Improving Global Outcomes (KDIGO)** defines six eGFR categories, cross-classified by albumin-to-creatinine ratio (ACR), with **substantial prognostic implications** for treatment and survival.
- **G5 (eGFR <15 mL/min/1.73 m²) is end-stage kidney disease (ESKD)**, also termed end-stage renal disease (ESRD), requiring dialysis or transplantation.
- **Morbidity and mortality rise with CKD stage, driven mainly by cardiovascular disease** (Table 91.2).
- **ESRD incidence is falling slightly but consistently** (Fig. 91.1), yet **demand far exceeds supply**: about 40,000 patients join the waiting list yearly against about 27,000 transplants.

TABLE 91.1 NUMBER OF PERSONS BY EGFR AND ACR

eGFR category	eGFR (mL/min/1.73 m ²)	Label	Distribution across ACR categories
G1	≥90	Normal or high	Largest group; overwhelmingly A1
G2	60–89	Mildly decreased	Second-largest; mostly A1
G3a	45–59	Mildly to moderately decreased	Mostly A1; A2 and A3 share rising
G3b	30–44	Moderately to severely decreased	A2 and A3 now a substantial fraction
G4	15–29	Severely decreased	A3 the largest subgroup
G5	<15	Kidney failure	Almost entirely A3; none in A1

ACR categories: **A1** normal to mildly increased (<30 mg/g); **A2** moderately increased (30–299 mg/g); **A3** severely increased (≥300 mg/g).

- **What the table is for:** the 14% prevalence is everyone outside G1–G2 with A1 — CKD is defined by **either** axis, and albuminuria rises as eGFR falls.
- **Trap:** normal eGFR with A2 or A3 albuminuria is CKD; much of CKD sits in the early categories.

TABLE 91.2 DEFINITION, CLASSIFICATION, AND PROGNOSIS OF CHRONIC KIDNEY DISEASE

eGFR (mL/min/1.73 m ²)	ACR <10	ACR 10–29	ACR 30–299	ACR ≥300
≥105	1.1	1.5	2.2	5
90–105	Reference	1.4	1.5	3.1
75–90	1	1.3	1.7	2.3
60–75	1	1.4	1.8	2.7
45–60	1.3	1.7	2.2	3.6
30–45	1.9	2.3	3.3	4.9
15–30	5.3	3.6	4.7	6.6

Relative risk of all-cause mortality by eGFR and ACR (mg/g). Source: Levey AS, de Jong PE, Coresh J, et al.: KDIGO Controversies Conference report, Kidney Int 80(1):17–28, 2011.

- **What the table is for:** eGFR and albuminuria are **independent, additive predictors of death** — risk rises down the rows and across the columns.
- **Pattern:** ACR ≥300 mg/g multiplies risk at every eGFR, even when eGFR is normal.

- **Trap:** the curve is not monotonic at the top — **eGFR ≥ 105 carries more risk than the reference band**, so a very high eGFR is not reassuring.

Fig. 91.1 — Incidence of ESRD in the US, 2000–2020 (cases per million): gradual decline across the period.

Renal Replacement Therapy

- **ESRD needs dialysis or transplantation** to maintain homeostasis and remove metabolic waste.

PERITONEAL DIALYSIS VERSUS HAEMODIALYSIS

	Peritoneal dialysis (PD)	Haemodialysis (HD)
Exchange membrane	Native peritoneum ; solute diffusion, water removed by altering instilled fluid concentration (glucose or icodextrin)	Synthetic semipermeable membrane ; diffusion for solutes, hydrostatic pressure gradient for ultrafiltration
Access	Silastic catheter placed laparoscopically; omentum tacked away to prevent malfunction	Large central venous catheter or arteriovenous (AV) fistula/graft ; blood flow 250–350 mL/min
Setting and schedule	Home, by patient, with or without a cycling machine; continuous	Centre or home; 3–5 hours, 3–5 times a week
Haemodynamic effect	Milder ; greater independence	Faster solute and fluid shifts → nausea, cramps, fatigue
Favoured in	Children, cardiovascular instability, no vascular access ; use increasing	—
Not feasible	Extensive abdominal surgery, peritoneal scarring	—
Complications	Catheter malfunction, hyperglycaemia, weight gain, hernia, tunnel infection, bacterial and fungal peritonitis	Access bleeding, graft infection, stenosis, thrombosis, ischaemic steal; central catheters carry higher haematogenous infection risk
Long-term failure	Peritoneal fibrosis → membrane loses efficiency	—

- **Permanent AV access is created well in advance** of anticipated dialysis.
- **Long-term dialysis complications:** cardiovascular disease (ventricular hypertrophy, coronary artery disease (CAD)), hyperphosphataemia, altered bone metabolism, progressive vascular calcification, calciphylaxis — and these **increase post-transplant morbidity and mortality**.
- **Transplantation is the optimal therapy for ESRD** — significantly better than dialysis, though inferior to the general population — but modality is individualised after weighing surgical and immunosuppression risk.

Concept — Every year on dialysis accrues cardiovascular and mineral–bone damage that the eventual transplant inherits; the modality that shortens dialysis exposure therefore wins, and **pre-emptive transplantation** is the logical extreme of that principle.

Kidney Transplantation Epidemiology

- **Transplant numbers rise yearly** (about 27,000 in 2023), **driven by deceased donation**; living donation is stable (Fig. 91.2) but **fell with COVID-19 in 2020**, recovering only partially by 2023.
- **Over 139,000 candidates were listed** in the US (2020 report of the Organ Procurement and Transplantation Network (OPTN) and Scientific Registry of Transplant Recipients (SRTR)).
- **Allocation revisions improved equity** of deceased-donor access once listed; **disparities in access to the list persist**, mainly socioeconomic.

- **A successful transplant may triple life expectancy** compared with listed patients who remain on dialysis.
- **First successful kidney transplant: identical twins, 1954.** Immunosuppression has since brought **1-year graft survival close to 100%**; **long-term graft survival and organ shortage** remain unsolved.
- **Quality of life improves**, and transplantation is **cost-effective**: after year one, annual care costs (drugs, tests, clinic visits) are significantly below dialysis.

PRE-EMPTIVE TRANSPLANTATION — TWO FIGURES IN THE SOURCE

Figure	Denominator	Value
Patients transplanted pre-emptively; the rest dialysed while awaiting transplant	Patients (denominator not specified)	3.1%
Transplants performed pre-emptively, 2021 (OPTN/SRTR)	Transplants performed	28.7% overall — living donor 51%, deceased donor 21%; paediatric recipients 56%

- **Different denominators and years**; the stable message: **pre-emptive transplantation is far commoner with a living donor and in children.**

Fig. 91.2 — Total kidney transplants by donor type, 2008–2020 (OPTN/SRTR 2020 Annual Data Report): deceased-donor transplants rising, living-donor transplants flat.

Concept — The benefit of transplantation is **front-loaded in cost and risk and back-loaded in survival and savings**: it costs and risks more in the first year, then outperforms dialysis thereafter.

Recipient Evaluation for Transplantation

- **Comprehensive evaluation**: ESRD carries cardiac, vascular and diabetic comorbidity and higher infection and malignancy risk.
- **Transplantation wins long-term but carries a higher risk of death in the first 3–6 months**, depending on recipient and donor factors (Fig. 91.3).

Fig. 91.3 — Landmark: Wolfe et al., *N Engl J Med* 1999. Adjusted relative risk of death in first deceased-donor transplant recipients versus wait-listed dialysis patients.

- **Relative risk 2.84 immediately after transplant → risk equal at 106 days → cumulative survival equal at 244 days → long-term relative risk 0.32.**

Timing of evaluation and listing

- **Timing** follows the pace of decline, symptoms and donor availability; **ideally evaluate before dialysis.**
- **Deceased-donor listing permitted at eGFR ≤ 20 mL/min/1.73 m².**
- **Pre-emptive transplantation is preferable** when a living donor is available or deceased-donor transplantation is accessible.
- **Elderly patients may gain from a faster elderly-donor kidney** over waiting for a standard kidney.

Aetiology and recurrence

- **Hypertension and diabetes are the commonest adult causes**; control them and modify lifestyle, including weight loss, before transplant to prevent recurrence in the graft.

RECURRENCE RISK OF THE PRIMARY DISEASE IN THE ALLOGRAFT

Recurrence risk	Diseases	Implication
Very low or none	Genetic disease: autosomal dominant polycystic kidney disease (ADPKD), congenital kidney dysplasia, Alport syndrome spectrum, genetic focal segmental glomerulosclerosis (FSGS)	No specific recurrence measures

Recurrence risk	Diseases	Implication
Higher	Primary FSGS , membranoproliferative glomerulonephritis (MPGN), membranous nephropathy	Pre-transplant counselling; monitor for recurrence after transplant
Histological often, clinical rarely	Immunoglobulin A (IgA) nephropathy	Impact on long-term survival less clear
Metabolic	Primary hyperoxaluria	May require combined liver and kidney transplant

- **Pattern:** disease intrinsic to the native kidney's genes cannot recur in a graft carrying donor genes; disease driven by circulating or immune factors can.
- **Trap:** FSGS appears on both sides — **genetic FSGS does not recur; primary FSGS does.**

Cardiac evaluation

- **CAD is highly prevalent in ESRD and the major cause of morbidity and mortality;** symptomatic patients are evaluated per current guidelines.
- **High-risk asymptomatic candidates** (diabetes, family history, smoking, prior CAD, poor functional reserve, long dialysis) → echocardiography, exercise or pharmacological stress testing, selectively coronary angiography.
- **Optimal screening is unclear;** studies challenge angiography in asymptomatic patients. Programmes follow local algorithms tied to waiting times and **retest long waiters.**

Vascular evaluation

- **Peripheral arterial disease raises risk** of limb complications, cardiovascular events and death; assess peripheral and cerebral vessels and anastomotic targets.
- **Heavy iliac calcification may preclude a safe anastomosis.**
- **Claudication may worsen after transplant** — counsel. **Amputation is not a contraindication** but marks higher risk.
- **Tests:** lower-limb and carotid Doppler; **non-contrast computed tomography (CT)** for calcification in long-standing diabetes, smokers and the elderly.
- **Non-healing foot ulcers and osteomyelitis** → not a candidate until healed.
- **Prior stents or bypasses** → define safe anastomotic targets with vascular surgery.
- **ADPKD with a family history of intracranial bleeding** → evaluate for cerebral aneurysm and manage before transplant.

Wound healing, obesity and frailty

- **Manage before transplant:** obesity, malnutrition, prior abdominal surgery, hernias, tobacco.
- **Obesity is a relative contraindication** — more wound complications, delayed graft function (DGF), acute rejection, possibly death. Body mass index (BMI) cut-offs are centre- and comorbidity-specific; pre-transplant weight interventions are proposed.
- **Frailty predicts mortality and morbidity;** assess and optimise, though strategies are unclear.

Haematological and psychosocial

- **Screen for haematological malignancy and bleeding or thrombotic disorders,** including liver disease and anticoagulants, for perioperative planning.
- **Psychosocial assessment for all. Unstable psychiatric disorder, non-adherence or substance abuse** must be treated and stable, with a support network, first.

Infection

- **Infection risk is higher** (diabetes, immunosuppressants). **Treat bacterial infection first**; dialysis catheter infection may cause sepsis or peritonitis and need catheter removal. Drug-resistant organism surveillance per local practice.
- **Mandatory screening**: tuberculosis (TB), hepatitis B and C, human immunodeficiency virus (HIV) → therapeutic, prophylactic or suppressive treatment. **Active TB is a contraindication until adequately treated.**
- **Serology for cytomegalovirus (CMV), herpes simplex virus (HSV) and Epstein-Barr virus (EBV)** defines reactivation risk and directs post-transplant chemoprophylaxis.
- **Vaccinate before transplant**: responses are better, and **live vaccines cannot be used after transplant.**
- **Chronic hepatitis** → assess fibrosis, screen for hepatocellular carcinoma (HCC), suppress virus. **Decompensated cirrhosis or severe portal hypertension** → consider simultaneous liver-kidney transplant.
- **Hepatitis C virus (HCV)**: direct-acting antivirals give near-universal sustained virological response, **allowing HCV-viraemic donor kidneys into HCV-negative recipients** with treatment — expanding the pool.
- **Hepatitis B virus (HBV)**: **vaccinate non-immune candidates**; positive serology → assess viraemia and liver damage, HCC surveillance, reactivation monitoring, continued antivirals.

FIG. 91.4 ALGORITHM FOR THE EVALUATION OF RENAL TRANSPLANTATION CANDIDATES

Component	Content (★ = area of special interest for urologists)
History and examination	Problem list★, medication list★, medical history★, immunisation history, physical examination★
Investigations	Chest radiograph, electrocardiogram★, abdominal ultrasound★, full blood count, chemistry screen★, international normalised ratio and activated partial thromboplastin time, infectious serology, prostate-specific antigen★, urinalysis ± culture★, histocompatibility testing
Screening	Colon and cervical cancer screening; mammogram (women ≥40 years); eye examination (diabetics); dental examination
Consults and support	Social services, dietetics, patient and family education, financial counselling
Sequential gates	Prior substance abuse → drug screens · morbid obesity → weight loss · non-adherence → contract · high risk of heart disease → evaluate
Exclusion gates	Active infection★, active malignancy★, high probability of perioperative mortality, unsuitable conditions for technical success★, non-compliance → if present, correct and re-evaluate; if uncorrectable, unacceptable

Modified from Barry JM: Current status of renal transplantation: patient evaluations and outcomes, Urol Clin North Am 28:788, 2001.

- **What the algorithm is for**: fixed work-up, then gates; each barrier is **correctable or a reason to decline.**

Concept — *Evaluation asks two questions* — **will the patient survive the operation, and will immunosuppression unmask something dormant** — *and almost every test on the list answers one of them.*

Malignancy Screening

- **Age-appropriate cancer screening is essential**: immunosuppression raises recurrence risk.
- **Waiting period** depends on type, stage, metastatic potential, immunogenicity and recurrence risk; **decisions in remission are collaborative** (oncology, transplant nephrology, patient, carers), aided by per-tumour consensus recommendations.

MALIGNANCY AND TRANSPLANT ELIGIBILITY

Category	Malignancies
Transplant possible without prior treatment (low grade)	Prostate cancer Gleason score ≤ 6 ; superficial non-melanoma skin cancer; incidental renal tumour <1 cm
No waiting period after curative treatment	Non-metastatic basal cell and squamous cell skin cancer; melanoma in situ; renal cell carcinoma <3 cm ; prostate cancer Gleason score ≤ 6 ; thyroid cancer (follicular or papillary <2 cm, low-grade histology); superficial bladder cancer
All other active malignancy	Transplant only after adequate treatment, with an individualised waiting period

- **Trap:** the two renal thresholds differ — **<1 cm incidental tumour needs no treatment first; <3 cm renal cell carcinoma needs no wait after cure.**

Additional Surgical Considerations

- **Anatomical evaluation** defines approach, site, targets and vessel integrity — especially after previous transplant, surgery or radiation.
- **Prolonged femoral dialysis access** (catheter or fistula) → exclude venous obstruction from stenosis or thrombosis. **A functioning lower-limb fistula causes venous hypertension and poor graft function.**
- **ADPKD:** chronic infection, cyst rupture, pain, haematuria, tumours and mass effect (early satiety, weight loss); **nephrectomy may be needed to make room** — before or simultaneously with transplant, weighing risks and benefits.
- **Other indications for pre-transplant native nephrectomy:** chronically infected kidneys, recurrent urinary tract infection (UTI) with reflux, significant proteinuria.
- **Heart, lung or liver candidates with renal dysfunction:** add a kidney only if disease is advanced or recovery unlikely; an OPTN **safety net** prioritises later kidney allocation if function does not recover.

ABO and Human Leukocyte Antigen

- **ABO compatibility is standard:** type O universal donor, type AB universal recipient.
- **Non-A1 donors (for example A2) can donate to B and O recipients** if the recipient's anti-A titre is acceptable per centre policy.
- **Human leukocyte antigen (HLA) compatibility means a negative crossmatch** between donor lymphocytes and recipient serum.
- **ABO- or HLA-incompatible transplant without preconditioning** → accelerated rejection, thrombosis, graft loss. **ABO-incompatible transplant is possible** with more intensive immunosuppression and rigorous monitoring.
- **The major histocompatibility complex (MHC), known as HLA,** consists of cell-surface glycoproteins.

HLA CLASS I VERSUS CLASS II

	Class I	Class II
Loci	HLA-A, HLA-B, HLA-C	HLA-DR, HLA-DQ, HLA-DP
Expression	All nucleated cells	Antigen-presenting cells and endothelial cells only

- **Sensitisation:** anti-HLA antibodies from transfusion, pregnancy, previous transplant and other immune stimulation (for example viral infection). **Preformed antibodies** → severe acute rejection and graft loss.

- **Detection is by solid-phase assay, mainly Luminex single-antigen beads;** defining each recipient's antibody repertoire allows a **virtual crossmatch**, expediting allocation.

Concept — Every barrier in this section is a **preformed antibody** — anti-A/B isoagglutinin or anti-HLA — and titres, crossmatch and bead assays are all ways of detecting it before the kidney meets it.

Kidney Donation

- **Living or deceased;** deceased donation is **donation after brain death (DBD) or donation after circulatory death (DCD)**.
- **Before 1980 all deceased donation was DCD.** The **Uniform Determination of Death Act, 1980** defined death as irreversible cessation of circulatory and respiratory function **or** of all functions of the entire brain, including the brainstem.

Deceased Donation

- **National Organ Transplant Act, 1984 (NOTA)** established the OPTN, which coordinates transplant centres and organ procurement organisations (OPOs); the **SRTR provides analytical support** for monitoring and policy.
- **Donor data generate a match run** on the national list; **allocation policy revised 2014, amended 2021.**
- **Ranking:** waiting time (from dialysis start or pre-emptive listing); biological criteria (HLA matching, sensitisation); **longevity matching by Kidney Donor Profile Index (KDPI) and Estimated Post-Transplant Survival (EPTS);** priority for centres **within 250 nautical miles** of the donor hospital.
- **Donor evaluation:** diagnosis and cause of death, neurological and cardiorespiratory status, medical and social history.
- **Donor contraindications:** active metastatic cancer, haematological malignancy, **history of melanoma**, untreatable viral illness.
- **Adequately treated bacterial infection, including bacterial meningitis, is acceptable;** viral meningitis or encephalitis needs careful assessment of transmission risk.
- **Laboratory:** ABO and HLA typing; screening for HIV, HBV, HCV, EBV, CMV, syphilis and toxoplasma by serology and **nucleic acid testing (NAT).**
- **Paediatric donors under 15 kg** → en bloc, with graft survival comparable to adult transplants.

DONATION AFTER BRAIN DEATH VERSUS DONATION AFTER CIRCULATORY DEATH

	DBD	DCD
Circulation at procurement	Maintained , with adequate oxygenation	Life support withdrawn; organs recovered after declaration of death
Ischaemic injury	Minimal	Warm ischaemia → expedient cannulation and cooling essential
Procurement	Standard sequence (below)	Rapid access to aorta and inferior vena cava (IVC) for perfusion and venting; then as DBD
Preservation	Static cold storage	Usually hypothermic machine perfusion to reduce DGF

Deceased-Donor Procurement

Long midline chest and abdominal incision with sternotomy → three exposure goals: (1) mobilise small- and large-bowel mesentery off the retroperitoneum; (2) expose and control aorta and IVC; (3) locate sites for cannulation, perfusion and venting → **recognise aberrant renal vessels; preserve accessory arteries arising from the iliacs before cannulation** → open Gerota's fascia laterally for ice cooling during perfusion → if liver and pancreas are procured, dissect them, perfuse and remove them → mobilise kidneys off the posterior

wall **with perinephric fat, preserving periureteric tissue down to the pelvis** → divide ureters → aorta and IVC: upper limit set by liver procurement, divided inferiorly at the aortic bifurcation → retract cut ends, kidneys and ureters anteriorly; divide prevertebral tissue → **remove both kidneys en bloc** → back table: separate, identify vessels, excise excess perinephric tissue, document anatomy, package for **static cold storage**.

Kidney Cold Preservation and Machine Perfusion

- **Immediate cooling to 4 °C** sharply reduces metabolism and damage; **preservation fluid maintains intracellular homeostasis**, limiting anaerobic glycolysis and acidosis; **reperfusion free radicals** compound the injury.
- **Common fluids: University of Wisconsin (UW) solution and histidine-tryptophan-ketoglutarate (HTK) solution.**
- **Pulsatile (hypothermic machine) perfusion** extends preservation time and **reduces DGF, with no difference in long-term graft survival.**
- **Normothermic perfusion:** limited clinical experience.

Concept — *Preservation buys time by slowing metabolism, not stopping it; every measure here targets ischaemia-reperfusion injury, which shows clinically as DGF — hence machine perfusion improves DGF but not long-term survival.*

Living Kidney Donation

- **Advantages:** significantly better graft and patient survival; recipient optimised for a planned operation; pre-emptive transplant possible; **negligible risk of donor-transmitted infection or malignancy.**
- **Multidisciplinary team:** independent donor advocate, social worker, physician, surgeon, coordinators.
- **Exclusions:** concern for kidney disease, impaired function, malignancy history, chronic or active infection, cardiovascular disease, diabetes, significant obesity.

Evaluation sequence: self-administered questionnaire (medical, social, family history) → physical and psychosocial evaluation, blood group and HLA typing → **renal function by at least two methods, including a measured technique** → comprehensive metabolic panel including glycated haemoglobin (HbA1c), lipid and coagulation profiles → serology per OPTN, endemic infections, age-appropriate cancer screening → **anatomy by CT angiography (CTA) or magnetic resonance angiography**, split function if needed → cardiac evaluation in selected cases.

- **Urological evaluation:** stone disease, genitourinary malignancy, recurrent UTI, previous urological surgery, bladder outflow symptoms. **Metabolic stone work-up** if stone history; **bilateral stones generally exclude.**
- **Counselling:** healthy lifestyle, ideal weight, regular care — **stricter in younger donors.**

LIVING DONOR RISKS FOR COUNSELLING

Risk	Figure
Long-term kidney function	60–80% of pre-donation level
Lifetime ESRD risk (Muzaale et al., 2014)	Donors 90 per 10,000 · matched healthy non-donors 14 per 10,000 · unscreened general population 326 per 10,000
Perioperative complications	About 16%, mostly gastrointestinal
Clavien grade ≥IV	2.5%
Mortality	3 per 10,000
Laparoscopic-to-open conversion	0.26–3.34%

Risk	Figure
Pregnancy after donation	More gestational hypertension and pre-eclampsia; no increase in maternal or fetal complications

- **Trap:** donors' ESRD risk is lower than the general population's but **higher than matched healthy non-donors — the correct comparator.**

Living Donor Nephrectomy

- **Donor safety is the primary goal** at every stage.
- **Laparoscopic donor nephrectomy first described by Ratner, 1995;** now the preferred approach. **Two-thirds are hand-assisted laparoscopic; in the US fewer than 2% are open.**
- **Minimally invasive nephrectomy is the procedure of choice:** faster recovery, less pain, higher acceptance, better cosmesis.
- **Open nephrectomy** is reserved for complex anatomy or previous surgery and is **more often needed on the right.**
- **Donor-specific technique:** minimal handling of the kidney; careful vascular dissection for adequate length **without compromising donor safety; preserve periureteric tissue** to avoid ureteric ischaemia; **minimise warm ischaemia** — divide ureter and vessels and extract expediently.

Kidney Paired Donation

- **Incompatibility arises from ABO mismatch or donor-specific antibodies (DSA)** from sensitising events (transfusion, pregnancy, previous transplant).
- **Incompatible pairs are matched with other pairs** through exchange databases (National Kidney Registry, Alliance for Paired Donation, United Network for Organ Sharing), maximising transplant chances.

Concept — *Living donation is an operation on a healthy person for another's benefit, so every decision — selection, counselling figures, surgical approach — is judged first by donor harm and only second by graft quality.*

KIDNEY TRANSPLANTATION SURGICAL PROCEDURE

Preparation for the Transplantation

- **On admission,** re-review suitability, dialysis need and interval health changes.
- **Counsel on DGF,** primary non-function, arterial and venous thrombosis, bleeding, urinary complications, immunosuppression side effects, and the **rare transmission of donor-derived infection or malignancy.**
- **Antibiotic prophylaxis** covers skin flora and urinary contaminants or colonisation, guided by previous positive cultures.
- **Safety check including ABO verification** before induction of anaesthesia.
- **Central venous catheter (CVC)** for drugs and sampling; multiple previous dialysis lines with possible stenosis → **pre-operative interventional radiology placement.**
- **Foley catheter after induction,** accessible intraoperatively for **bladder distension** during the ureteric anastomosis.

Preparation of the Kidney Allograft

PERFUSATE BY COLD ISCHAEMIA TIME

Cold ischaemia	Setting	Perfusate
Very short	Living donor, donor and recipient operations simultaneous or overlapping	Cold lactated Ringer solution
Longer	Sequential transplantation, exchange programmes	UW or HTK solution

- **Verify graft anatomy:** number of vessels and ureteric anomalies.
- **Renal vein:** secure all tributaries — suprarenal, lumbar, gonadal; **the right gonadal vein only rarely drains into the renal vein.**
- **Ligate perivascular neurolymphatic tissue** to prevent lymphocele.
- **Handle the artery gently** to avoid intimal injury.
- **Preserve periureteric tissue; do not dissect in the angle between the lower pole and the ureter.**
- **Excise excess perinephric fat;** in older donors remove adherent fat carefully to **avoid decapsulation.**
- **Pre-perfusion wedge biopsy** (authors' routine) records baseline histology — useful in delayed function and for counselling living donors about abnormalities.

Deceased-donor allografts

- **Same preparation, but vessels are long and unidentified polar vessels may be present.**
- **Right kidney: the accompanying IVC can extend the short right renal vein** (Fig. 91.5).
- **Vascular stapler** closure allows the vein to be shortened at the time of anastomosis, unlike a suture line.
- **Two renal veins of equal size should ideally both be preserved;** otherwise the smaller can be ligated.
- **Artery:** clear perivascular tissue; if the **Carrel patch is atherosclerotic** and unusable, use the renal artery alone as in living-donor transplant.

Fig. 91.5 — Methods of extending the right renal vein (A–G): modifications of the IVC and a free graft of donor external iliac vein. A and B are valuable when the cephalad right renal vein has been compromised by separating the liver graft. Sources: A, B patch and flap techniques (Barry and Lemmers, 1995); C (Barry and Fuchs, 1978); D, F clam-shell technique (Barry, Hefty and Sasaki, 1988); E lengthening technique (Corry and Kelley, 1978); G spiral gonadal vein graft (Nghiem, 1989).

Multiple renal arteries

- **Incidence 18–43%, bilateral in 15%.** A meta-analysis found **more complications and DGF but comparable long-term graft and patient survival.**
- **Arteries close together** → longer Carrel patch; if of adequate size, shorten the patch, or reimplant the artery into the patch nearer the main artery or into the main artery itself (Figs. 91.6, 91.7).
- **The lower pole artery must be preserved — it supplies the ureter. Upper pole branches ≤ 2 mm may be ligated** if reconstruction is complex.

RECONSTRUCTION TECHNIQUES FOR MULTIPLE ARTERIES

Technique	Use
Conjoined side-to-side	Living donors: performed on the back table before implantation; may be mounted on a Gore-Tex or bovine pericardial patch depending on recipient iliac quality (Fig. 91.8)
End-to-side, smaller into larger	Unequal arteries
Two separate anastomoses	To the external iliac artery

Technique	Use
One artery to internal iliac	Depending on nature and size of the artery

Choice of Site

- **Factors:** iliac artery quality, iliac vein patency, failed graft or scar, stoma, space in ADPKD, groin dialysis access.
- **Right iliac fossa is the traditional choice:** the common iliac vein can be mobilised by securing internal iliac tributaries when the renal vein is short — **especially relevant for living-donor grafts.**
- **Contralateral placement is suggested:** the renal pelvis and ureter then lie **anterior**, easing future urinary reconstruction.
- **Retroperitoneal placement is ideal.**

GIBSON VERSUS HOCKEY STICK INCISION

	Gibson incision	Pararectal J-shaped hockey stick incision
Course	Oblique, from 2 cm above the pubic symphysis , laterally to 2–3 cm superomedial to the anterior superior iliac spine (ASIS) ; extendable superiorly	From the midline above the symphysis, superolaterally along the rectus sheath edge
Length	Shorter , mostly below the umbilical waistline	Longer
Access	Iliac vessels, retroperitoneal	Better access to common iliac vessels, IVC and intraperitoneum
Incisional hernia	Lower	Higher , with denervation of lower rectus

- **Gibson pitfalls:** incision over the symphysis → painful scar; near the ASIS → injury to the **lateral cutaneous nerve of the thigh**.

Gibson exposure: split external oblique aponeurosis and rectus sheath → retract lateral rectus edge medially → enter retroperitoneal fat plane → secure and divide **inferior epigastric vessels** (unless used for a lower pole artery anastomosis — rarely feasible in elderly and diabetic patients because of calcification) → free, preserve and retract the spermatic cord; **secure and divide the round ligament in women** → incise transversalis fascia → sweep peritoneum superomedially → divide internal oblique and transversus as needed.

Recipient vessel exposure: expose psoas and iliac vessels → self-retaining retractor with the **lower blade clear of the femoral nerve** (suture retraction in thin patients) → **ligate all perivascular lymphatics** (lymphocele prevention); avoid the genitofemoral nerve → expose enough length for safe clamping; mobilise common and internal iliac arteries for multiple arteries; **divide the internal iliac vein to mobilise the common iliac vein** for a short renal vein, securing tributaries → place graft in the retroperitoneal pocket to mark anastomotic targets.

- **A tortuous, medially lying iliac artery with a laterally placed vein** helps with a short renal vein.
- **Targets:** external iliac artery and vein (commonest); common iliac artery and IVC; rarely aorta and IVC.

Vascular anastomoses: clamp iliac vein → venotomy; **resect valves** carefully → graft brought in wrapped in cold gauze → **confirm polar orientation** → renal vein end-to-side, monofilament polypropylene 5-0/6-0 → clamp iliac artery **avoiding calcified spots** → arteriotomy (aortic punch for living donors) → flush with heparinised saline → artery end-to-side, polypropylene 5-0/6-0 (**interrupted for small living-donor vessels**)

→ bulldog clamps on graft vessels, remove iliac clamps, check haemostasis → remove bulldogs to reperfuse, check haemostasis again.

- **Venous anastomosis first**, to minimise arterial ischaemia time.
- **Atherosclerotic recipient arteries** (especially with living-donor kidneys) → implant the graft artery on a **Gore-Tex or bovine pericardial patch** on the back table for a wider anastomosis (Fig. 91.8).
- **Bleeding needing re-clamping after reperfusion** → proximal arterial clamp; the kidney stays perfused by backflow.
- **Position the kidney** without stretch or kinking; confirm proximal and distal recipient vessel integrity.
- **Hume test**: finger occlusion of the renal vein → pulsatile swelling of the kidney, promptly relieved on release.

Ureteric anastomosis

- **Commonest technique: extravesical ureteroneocystostomy (Lich, 1961)**; choice depends on ureteric vascularity and length and on recipient bladder quality.

Lich–Gregoir sequence: fill bladder via catheter → dissect perivesical fat, clear detrusor → myotomy along the intended ureteric course → open mucosa at the distal end of the myotomy → trim ureter to length and spatulate → mucosal anastomosis with fine (6-0/5-0) absorbable suture (polyglyconate or polydioxanone), with or without a stent → close muscle over the mucosal suture → **short submucosal tunnel**.

Fig. 91.9 — Lich–Gregoir (Konnak et al., 1975): anterolateral seromuscular incision down to bulging mucosa; bladder drained, mucosa incised, ureter anastomosed with fine absorbable sutures; a distal anchoring stitch prevents proximal migration of the ureter in the tunnel; seromuscular layer closed loosely over the ureter.

- **Alternatives**: ureteroureterostomy, pyeloureterostomy, pyelovesicostomy, anastomosis to ileal conduit or pouch.
- **Double ureters** → conjoined (**Wallace**) or individual anastomosis.
- **Document the stent and a retrieval plan**: forgotten stents cause calcification, graft dysfunction, even loss.
- **Drain**: surgeon's preference, by extent of dissection and lymphatic control; **closed system, removed early**.

Concept — The operation is ordered to protect the graft's two vulnerabilities, **ischaemia and the ureteric blood supply**: the vein goes first to shorten arterial ischaemia, the lower pole artery and periureteric tissue are untouchable, and contralateral placement keeps the ureter reachable for later repair.

Paediatric En Bloc Kidney Transplantation

- **Donors under 15 kg**: both kidneys transplanted en bloc with the **aorta and IVC as vascular conduits**; ureters anastomosed conjoined or individually (Fig. 91.10).
- **Ideal recipient**: adult, **BMI <30**, low immunological risk, low recurrence risk, no significant vascular disease.
- **Early vascular thrombosis and graft loss 10–15%**, yet **long-term graft survival equals living-donor transplantation**.

Surgical Complications

- **Early complications**: haematuria, urine leak, haematoma and haemorrhage, lymphocele, renal vein and artery thrombosis. Urological complications are covered in Chapter 92.
- **Bleeding** (brisk sanguineous drain output, falling haematocrit, graft swelling; risk raised by renal failure, antiplatelets, anticoagulants) → transfusion and **immediate exploration**. Large haematoma causes pain and **pressure on renal vein and bladder**. Ultrasound and CT help, especially if obese.

Vascular Complications

Intraoperative prevention

- **Clamp and anastomose clear of atheroma** — clamp injury causes intimal dissection, stenosis, **limb-threatening ischaemia**. Trial the final position to avoid stretch or kinking.
- **Renal artery usually to external or common iliac. Avoid the internal iliac artery if the contralateral one is occluded or already used** → gluteal claudication and erectile dysfunction.
- **Short right renal vein** → extend with donor IVC, or mobilise the recipient iliac vein by dividing the internal iliac vein.
- **Previous deep vein thrombosis (DVT), leg swelling or groin dialysis line** → image iliac vein patency. **Ipsilateral functioning groin AV fistula** may need ligation to avoid graft venous hypertension.
- **Resect venous valve cusps** at the anastomosis — inclusion in the suture line causes outflow obstruction.
- **On reperfusion, a flaccid soft kidney or a tense engorged kidney** → evaluate inflow and outflow, anastomoses, twisting or stretching of the pedicle; intraoperative ultrasound; occasionally **take-down and re-anastomosis**.

GRAFT ARTERIAL THROMBOSIS, VENOUS THROMBOSIS AND RENAL ARTERY STENOSIS

	Arterial thrombosis	Venous thrombosis	Renal artery stenosis
Incidence	0.2–5%	0.1–8.2%	1–23%
Presentation	Sudden anuria , usually immediately post-operative	Swollen, painful graft, gross haematuria , acute fall in urine output	Declining graft function, refractory hypertension , fluid retention, flash pulmonary oedema (renin–angiotensin–aldosterone activation)
Ultrasound	Absent Doppler signal	Reversed diastolic flow ; swollen graft; clot in renal vein	Raised velocity across stenosis; parvus-tardus waveform distally
Confirmation	Exploration	Exploration	Contrast imaging — gold standard
Management	Immediate ultrasound and exploration ; catheter thrombolysis described for short ischaemia; nephrectomy if unsalvageable	Immediate exploration : thrombectomy, intraoperative ultrasound, outflow revision; nephrectomy if necrotic	Percutaneous angioplasty or stenting — treatment of choice , reverses symptoms; surgical revision, interposition or bypass risks graft loss (Figs. 91.11, 91.12)

- **Prothrombotic patients** need perioperative anticoagulation, at the cost of more bleeding.
- **Multiple renal arteries may increase thrombosis risk. Loss of a lower pole artery** → infarction → ureteric ischaemia, necrosis or scarring → urine leak or hydronephrosis from stricture.
- **Thrombosis risk factors**: extreme donor or recipient age, **cold ischaemia >24 hours**, anastomotic technical problems, hypercoagulability accentuated by hypovolaemia and acute rejection.
- **Prevention**: minimise ischaemic time, careful technique, recognise hypercoagulable states (antithrombin III, protein C and protein S deficiency, factor V Leiden, prothrombin G20210A mutation, antiphospholipid antibodies), perioperative anticoagulation when required.

Other vascular and biopsy complications

- **Torsion**: rare; **intraperitoneal grafts only** (retransplantation, simultaneous pancreas–kidney transplant). Acute anuria and pain; **emergent ultrasound and exploration are the only chance of salvage**. **Prophylactic nephropexy** when placed intra-abdominally.
- **Pseudoaneurysm**: rare (<1%) but serious — limb ischaemia and death from rupture. **Technical or mycotic** (bacterial or fungal, from donor or native source). Usually asymptomatic, **mostly found incidentally** when

imaging graft dysfunction; may cause pain, fever, distal thromboembolism, bleeding, haematoma.

Treatment: **graft nephrectomy and endovascular or open repair, with prolonged antibiotics.**

- **Biopsy complications:** minor (microhaematuria, small AV fistula, small haematoma) are common and self-limiting; **major (bleeding needing transfusion or intervention) are low (1.9%).** Subcapsular haematoma may cause a **Page kidney**. Ultrasound and contrast CT diagnose. **Large AV fistula with steal → coil embolisation;** large compressive haematoma → surgical evacuation.

Lymphocele

- **Perinephric pseudocyst** of variable incidence.
- **Risk factors:** extensive dissection, retransplantation, obesity, anticoagulation, **mechanistic target of rapamycin (mTOR) inhibitors.**
- **Mostly asymptomatic;** large ones compress iliac vein (leg oedema), renal vein, ureter (hydronephrosis), bladder (frequency). Ultrasound shows it; contrast CT maps anatomy pre-intervention; **fine-needle aspiration distinguishes urinoma, seroma, abscess.**
- **Treat only if symptomatic:** aspiration with or without sclerotherapy, percutaneous drain, laparoscopic or open fenestration into the peritoneum. **Infected lymphocele needs adequate percutaneous or open drainage.**
- **Success is highest with laparoscopic fenestration,** then open surgery; sclerotherapy and drain placement are intermediate; **aspiration alone is lowest.**
- **Prevention:** meticulous ligation of perivascular and hilar lymphatics; a drain placed at transplant also reduces incidence.

Transplant nephrectomy

- **Most grafts failing within the first year need nephrectomy.** Indications: **graft intolerance syndrome** (graft swelling, haematuria, graft pain, non-specific inflammatory symptoms), infection, tumour.

EARLY VERSUS DELAYED TRANSPLANT NEPHRECTOMY

	Early (within 6–8 weeks)	Delayed (graft intolerance)
Plane	Extracapsular simple nephrectomy	Subcapsular: dissect under the capsule, suture the pedicle at the hilum
Exception	—	Tumour in the graft → extracapsular even late; more complex, risk of vascular and nerve injury

- **Graft intolerance syndrome:** a course of steroids to suppress oedema is preferred first.
- **Asymptomatic failed graft: nephrectomy is not indicated simply to allow immunosuppression withdrawal,** because sensitisation risk persists.
- **Morbidity:** complications 20–30%, mortality 0–11% — intraoperative bleeding, haematoma, infection and sepsis, lymphocele, DVT, vascular injury, **stump infection with major bleeding,** nerve injury.

Concept — *The presentation localises the lesion: no inflow gives anuria, blocked outflow gives a swollen tender graft with haematuria, reduced inflow gives renin-driven hypertension and fluid overload — and the more acute the loss of flow, the more the answer is exploration rather than imaging.*

Kidney Transplant Immune Monitoring for Rejection

- **Agents essentially unchanged for two decades;** individualisation limited to sensitisation, infection, previous rejection.
- **Registry practice:** over 90% receive **induction;** over 90% receive **tacrolimus plus mycophenolic acid** maintenance; **maintenance steroid use is rising,** now more than two-thirds.

- **Acute rejection has fallen from at least 60% in the precyclosporine era to below 10% today.**
- **Long-term outcomes have improved only modestly.** Leading hypothesis: **chronic antibody-mediated injury** against allograft antigens, **subclinical** until accumulated damage causes advanced dysfunction.
- **Rejection is classified clinically or histologically; biopsy is required for accurate diagnosis.**

CLINICAL TYPES OF REJECTION

Type	Definition	Note
Hyperacute	Preformed high-level DSA at transplant → immediate severe injury and failure	Practically eliminated by histocompatibility testing and crossmatching
Acute	Rejection with acute decline in graft function	—
Subclinical	Histological rejection without dysfunction (no fall in function, no proteinuria)	Detected by surveillance biopsy or biomarkers
Chronic	Chronic deterioration with histology of chronic changes superimposed on rejection	—

- **Banff classification** standardises histological diagnosis and severity grading, with prognostic value; it recognises **T-cell-mediated rejection (TCMR)** and **antibody-mediated rejection (AMR)**, with separate inflammation and chronicity grades.
- **Risk factors — prior or more intense immune activation:** HLA mismatch, DSA, **younger recipient and older donor**, ABO incompatibility, prolonged cold or warm ischaemia and DGF, **non-adherence**, repeat transplant, previous rejection.
- **Impact of acute rejection varies** with timing, severity and functional recovery after treatment.

Monitoring for Rejection

REJECTION MONITORING TOOLS

Tool	Detects	Strength	Limitation
Routine laboratory: eGFR from serum creatinine, urinalysis for cells and protein	Graft dysfunction	Universal baseline	Detects only once dysfunction exists
DSA monitoring	AMR; risk-stratifies dysfunction	De novo DSA strongly predicts reduced graft survival — linked to greater HLA mismatch, especially class II , and insufficient immunosuppression	AMR can occur without detectable DSA (non-HLA antibodies; testing not yet standardised)
Surveillance biopsy	Subclinical rejection	Histological diagnosis	Used by a decreasing minority : timing mismatch, sampling error, subjective early interpretation, time, cost, complication risk
Donor-derived cell-free DNA (dd-cfDNA)	Injury, especially AMR ; subclinical rejection	Negative predictive value >90% → avoids unnecessary biopsy; prognostic for function and DSA formation	—
Gene-expression panels	Subclinical rejection	—	—

Treatment of Rejection

- **Treatment is guided by histological type and grade**, the degree of dysfunction, and DSA or other injury markers (dd-cfDNA). **Management of borderline rejection is debated.**

TREATMENT OF T-CELL-MEDIATED VERSUS ANTIBODY-MEDIATED REJECTION

	TCMR	AMR
Standardisation	Established	Not well standardised: depends on severity, coexisting TCMR, DSA and complement activation, chronic changes
First line	Intravenous steroids	Combinations of glucocorticoids, intravenous immunoglobulin (IVIG) , plasmapheresis , rituximab ; eculizumab in selected patients
Escalation	Antithymocyte globulin (ATG) for Banff grade II or III	Newer: anti-interleukin-6 (IL-6) therapy , particularly for chronic active AMR

Concept — *Acute T-cell rejection has been largely conquered; the unsolved problem is **slow, subclinical antibody injury**, which is why monitoring has moved from creatinine towards DSA, dd-cfDNA and biopsy, and why AMR treatment remains unstandardised.*